



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 10	Team No- 9	Project Title- Multi-Vendor Campus Marketplace & Delivery Platform
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1. Abstract

Campus communities rely on multiple independent vendors, such as stationery shops, cafeterias, print centers, and grocery outlets, but students must currently interact with each vendor separately through informal channels like phone calls or in-person visits. This lack of a unified digital interface leads to long wait times, mismanaged orders, and missed revenue opportunities for small vendors who lack the resources to build individual e-commerce solutions. The proposed Multi-Vendor Campus Marketplace & Delivery Platform is an integrated web-based solution that allows multiple campus vendors to list and manage their products and services under a single platform, while students browse, compare, and place orders across vendors through one unified cart and checkout system. The platform provides role-based access for students, vendors, and delivery agents, along with real-time order tracking, secure payment integration, and a delivery-agent assignment module for handling pickup and drop-off within the campus. On the backend, the system follows a distributed, database-driven architecture that ensures data consistency across concurrent vendor transactions, scalable order processing, and efficient inventory synchronization. The proposed solution aims to streamline vendor operations, reduce order-fulfillment time, and give students a centralized, reliable, and convenient way to access campus services digitally, while also providing vendors with sales insights through an analytics dashboard.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090182	V. Bhardwaj	
2520090089	P. Uday	
2520090094	A. Chaitanya	

3. Frameworks / Platforms / Software Used

Frontend: React.js, HTML5, CSS3, Tailwind CSS. Backend: Node.js with Express.js (REST APIs). Database: MongoDB / MySQL for distributed data storage. Authentication: JWT-based role authentication for students, vendors, and delivery agents. Payments: Razorpay / Stripe payment gateway integration. Other tools: Git & GitHub for version control, Postman for API testing, and Firebase Cloud Messaging for real-time order notifications.

Guide Name and Signature (remarks if any)